

Continuous Distillation

Group 4

Jake Proctor, Milos Markovic,
Courteney Fabbri, Erica Caruso, Mike Simon
Chemical Engineering Lab II
5 March 2020

Introduction

- The main purpose of this lab was to study the separation process of a binary mixture (EtOH and H₂O)
- Separation method used was continuous distillation
- Performed distillation with two different feed positions (stage 1 and stage 4) – stripping only section and stripping/rectifying sections
- This lab also served to study the accuracy of the using theoretical formulas in calculating column efficiency

Theory

- McCabe-Thiele Assumptions
 - Each stage is at equilibrium
 - 100% efficiency of each stage
- Operating Equations

Top

$$y = \frac{L}{V}x + \left(1 - \frac{L}{V}\right)x_D$$

Bottom

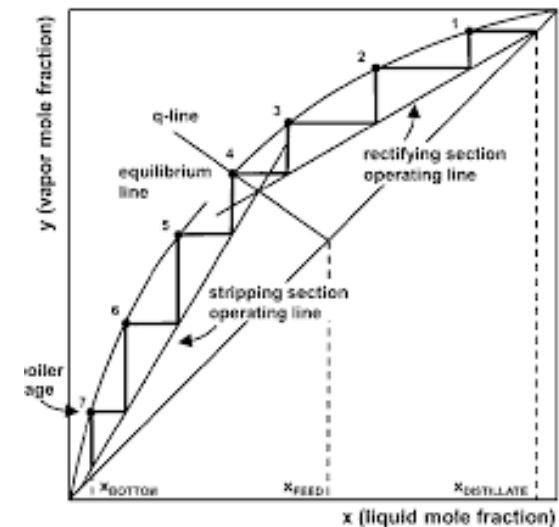
$$y = \frac{\bar{L}}{\bar{V}}x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$$

Feed

$$y = \left(\frac{q}{q-1}\right)x + \left(\frac{z}{1-q}\right)$$

- Column Efficiency

$$n + 1 = \frac{\log\left[\left(\frac{x_{EtOH}}{x_{H2O}}\right)_d \left(\frac{x_{H2O}}{x_{EtOH}}\right)_b\right]}{\log(\alpha_{EtOH/H2O})_{avg}} \quad \text{AND} \quad E = \frac{\text{number of theoretical plates}}{\text{number of actual plates}} * 100\%$$



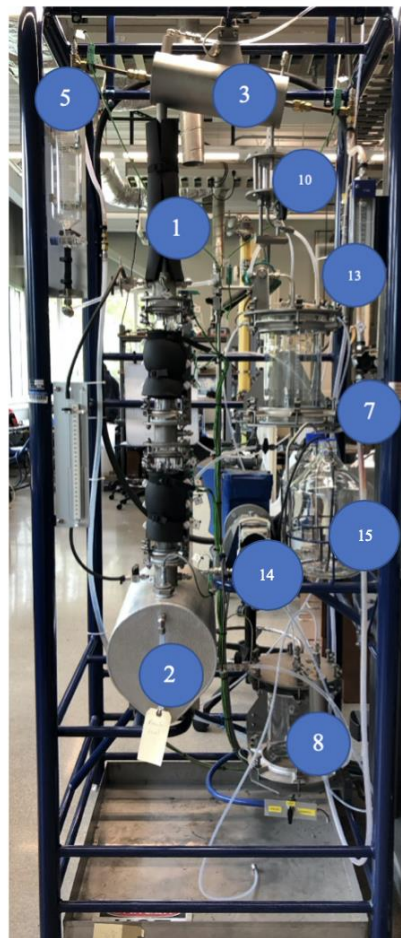
Apparatus

- Main Equipment

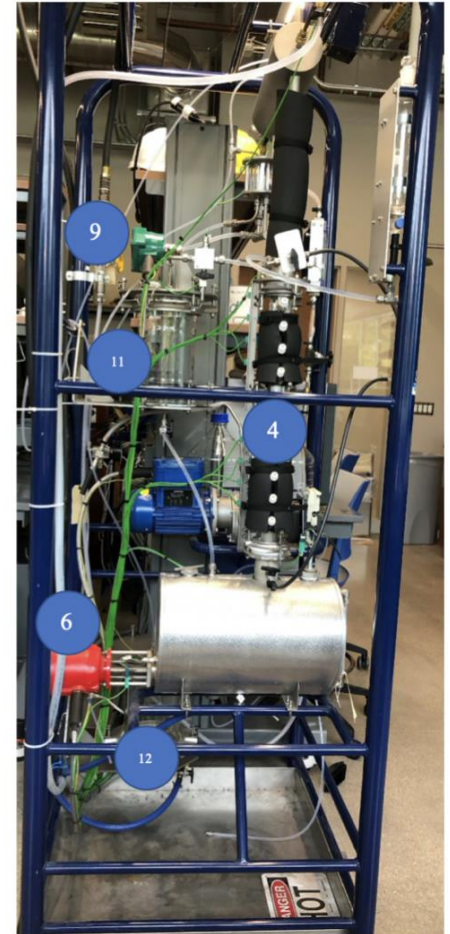
1. Column
2. Reboiler
3. Condenser
4. Trays
5. Manometer
6. Heater
7. Valves*
8. Heat exchanger
9. Reflux valve
10. Decanter
11. Top product receiver
12. Bottom product receiver
13. Cooling water flowmeter
14. Peristaltic feed pump
15. Feed holding jug

* Note there are many valves in this experiment

Front view:



Side view:



Material & Supplies

- Material list

- Ethanol
- Water
- Large graduated cylinder
- Measuring cylinder
- Timer
- Bottle
- Refractometer
- Test tubes
- Computer
- Armfield console
- Pipettes
- Tap water
- Water bath and rack
- Funnel



Procedure

- Prelab Steps
 1. Familiarize team with equipment
 2. Calculate volume for EtOH and H₂O for the two required solutions
 3. Create calibration graph for feed pump
- Lab Day Steps
 1. Make both solutions of EtOH and H₂O
 2. Load boiler with 10L solution and heat until constant temperatures on computer
 3. Check calibration curve for refractometer with 10 wt% increment composition mixtures of EtOH and H₂O from last semester

Part A

Start pumping feed into middle of column, set reflux ratio ratio to 5:1, reboiler power to 0.75kW and collect top and bottom samples for refractometer indexing

Part B

Move feed pumping location to top stage and repeat Part A steps

Experimental Challenges

- Difficulties
 - Understanding the operation of the column
- Things to watch out for during the experiment
 - Not to open valves too fast and overflow test tubes while collecting
 - Make sure the cooling water is on and at 3000 cc/min
 - Turn valves the correct direction
 - Careful not to damage the feed tubing in the pump
- Lessons Learned
 - Best to have one person taking data, one person performing refractometer indexing, and two/three people operating column
 - Use prelab time to calculate the flowrates of the feed pump

Safety

- Wear gloves and goggles at all times
- Boiler and samples collected are HOT – use caution and heat gloves
- Be careful not to overflow hot samples while collecting
- Careful when changing feed locations (tubing and fittings will be hot)